AFQMC in the Majorana Representation

> The aim of this note is to formulate the standard AFQMC algorithm in the *Majorana* basis, which is a general representation to deal with Hamiltonians with fermion-pairing terms. The Majorana QMC was first proposed in Phys. Rev. B **91**, 241117(R) (2015).

We consider a generic BdG Hamiltonian

$$H_{\mathrm{BdG}} = \sum_{ij} t_{ij} c_i^\dagger c_j + \frac{1}{2}\sum_{ij} \Delta_{ij} c_i^\dagger c_j^\dagger + \frac{1}{2}\sum_{ij} \Delta_{ij}^* c_i c_j. \tag{1}$$

To satisfy the anti-commutation relations of complex fermions, one requires the Δ to be skew-symmetric, i.e. $\Delta_{ij} = -\Delta_{ji}$. We ignore the spin index for convenience, which can however get restored afterwards without ambiguity.

To deal with a general BdG Hamiltonian in the framework of AFQMC, it's convenient to work in the Majorana basis. Consider the Majorana transformation,

$$c_i = \frac{1}{2}(\gamma_i^1 + i\gamma_i^2), \quad c_i^\dagger = \frac{1}{2}(\gamma_i^1 - i\gamma_i^2), \tag{2}$$

where the Majorana fermions γ_i^m with $m = \{1, 2\}$ also obey the anti-commutation relations $\{\gamma_i^m, \gamma_j^n\} = 2\delta_{mn}\delta_{ij}$, and this implies that $\gamma_i^m \gamma_i^m = 1$. The inversed transformation follows as $\gamma_i^1 = c_i + c_i^\dagger; \gamma_i^2 = -i(c_i - c_i^\dagger)$. We then introduce the Majorana spinor $\gamma = (\gamma_1^1, \cdots, \gamma_N^1, \gamma_1^2, \cdots, \gamma_N^2)^T$,which is related to the usual Nambu spinor $c = (c_1, \cdots, c_N, c_1^\dagger, \cdots, c_N^\dagger)^T$ via

$$\gamma = \begin{pmatrix} \mathbb{I} & \mathbb{I} \\ -i\mathbb{I} & i\mathbb{I} \end{pmatrix} c, \quad c = \frac{1}{2}\begin{pmatrix} \mathbb{I} & i\mathbb{I} \\ \mathbb{I} & -i\mathbb{I} \end{pmatrix} \gamma. \tag{3}$$

Under this basis transformation, we can formulate the H_{BdG} in the Majorana representation as

$$H_{\mathrm{BdG}} = \frac{1}{2}\overline{\gamma}\boldsymbol{h}\gamma + \frac{1}{2}\mathrm{Tr}[\boldsymbol{t}]; \quad \boldsymbol{h} = \frac{i}{2}\begin{pmatrix} \mathrm{Im}[\boldsymbol{t}] + \mathrm{Im}[\boldsymbol{\Delta}] & \mathrm{Re}[\boldsymbol{t}] - \mathrm{Re}[\boldsymbol{\Delta}] \\ -\mathrm{Re}[\boldsymbol{t}] - \mathrm{Re}[\boldsymbol{\Delta}] & \mathrm{Im}[\boldsymbol{t}] - \mathrm{Im}[\boldsymbol{\Delta}] \end{pmatrix}, \tag{4}$$

where $\overline{\gamma} = \gamma^T, \boldsymbol{t}_{ij} = t_{ij}$, $\boldsymbol{\Delta}_{ij} = \Delta_{ij}$. The hermiticity of H_{BdG} requires that $\boldsymbol{h}$ is skew-symmetric and purely complex. This can be proved by noting $\boldsymbol{T}^H = \boldsymbol{T}$ and $\boldsymbol{\Delta} = -\boldsymbol{\Delta}^T$.

Trace of the exponentials of Majorana bilinears [1]

Hereafter we will simply consider the bilinear Hamiltonian $h = \frac{1}{4}h_{ij}\gamma_i\gamma_j$ in terms of Majorana fermions with $h_{ij} = -h_{ji}$ and $0 \leqslant i, j < 2N$. The summations over the indices i and j are implicitly assumed. The first essential quantity we will encounter and calculate in the AFQMC is the trave of exponentials of fermion bilinears, e.g. $Z(h) \equiv \mathrm{Tr}[e^{\frac{1}{4}h_{ij}\gamma_i\gamma_j}]$, where the Tr is taken over the complete many-body Hilbert space of 2^{2N} dimensions in the case of Majorana representation.

A nice formula of $Z(h)$ is derived in *J. Stat. Mech.* (2014) P11006 in terms of beautiful algebras. It states that for a $2N$-dimensional, skew-symmetric matrix $h \in \mathrm{Skew}(2N, \mathbb{C})$, the following formula holds,

$$Z(h) \equiv \mathrm{Tr}\left[e^{\frac{1}{4}h_{ij}\gamma_i\gamma_j}\right] = \frac{\mathrm{Pf}\left(e^{-h} - e^{h}\right)}{\mathrm{Pf}\left(e^{-h/2} - e^{h/2}\right)}, \tag{5}$$

where Pf denotes the Pfaffian of skew-symmetric matrices. Besides, a simpler formula is possible which is correct up to a sign ambiguity $Z(h) = \sqrt{\det(1+e^h)}$. This simpler formula is adopted in the original paper of Majorana QMC. [2] It's straightforward to check that the square of the pfaffian version $Z(h)$ equals $\det(1+e^h)$, and the sign ambiguity is formally determined by the smoothness of $Z(h)$ in terms of small perturbations in h.

The above result can get easily extended to the case of products of exponentials of Majorana bilinears. Consider

$$Z\left(\{h(n)\}\right) = \mathrm{Tr}\left[\prod_n e^{\frac{1}{4}h(n)_{ij}\gamma_i\gamma_j}\right], \tag{6}$$

and note that the map $h \to \frac{1}{4}h_{ij}\gamma_i\gamma_j$ forms a representation of the Lie algebra of skew-symmetric matrices [1], we have

$$\prod_n e^{\frac{1}{4}h(n)_{ij}\gamma_i\gamma_j} = e^{\frac{1}{4}C[\{h(n)\}]_{ij}\gamma_i\gamma_j} \to \prod_n e^{h(n)} = e^{C[\{h(n)\}]}. \tag{7}$$

$C\left[\{h(n)\}\right]$ is given by a Baker Campbell Hausdorff (BCH) type series, and we assume it has in general a finite radius of convergence. Also note that $C\left[\{h(n)\}\right]$ remains to be skew-symmetric as long as all $h(n)$ are skew-symmetric, since the commutation preserves the skew-symmetric properties.

After these preliminary works, we can immediately write

$$Z\left(\{h(n)\}\right) = \mathrm{Tr}\left[\prod_n e^{\frac{1}{4}h(n)_{ij}\gamma_i\gamma_j}\right] = \mathrm{Tr}\left[e^{\frac{1}{4}h'_{ij}\gamma_i\gamma_j}\right] = \frac{\mathrm{Pf}\left(e^{-h'} - e^{h'}\right)}{\mathrm{Pf}\left(e^{-h'/2} - e^{h'/2}\right)}, \tag{8}$$

where $h' = \ln[\prod_n e^{h(n)}]$ is a series of commutators, denoted by $C\left[\{h(n)\}\right]$ above. The simpler formula in terms of the square root of matrix determinant now becomes

$$Z\left(\{h(n)\}\right) = \sqrt{\det\left(1+e^{h'}\right)} = \sqrt{\det\left(1+\prod_n e^{h(n)}\right)}, \tag{9}$$

where the sign ambiguity still remains.

Equal-time Majorana Green's functions

The ensemble averaged physical observables can be obtained within the framework of AFQMC given the configuration $\mathcal{C}$ of auxiliary fields as

$$\langle O\rangle_{\mathcal{C}} = \frac{\mathrm{Tr}\left[U_{\mathcal{C}}(\beta,\tau)OU_{\mathcal{C}}(\tau,0)\right]}{\mathrm{Tr}\left[U_{\mathcal{C}}(\beta,0)\right]}, \tag{10}$$

where the propagation operator $U_{\mathcal{C}}(\tau_1,\tau_2) = \prod_{\tau=\tau_2}^{\tau_1} e^{-\Delta_\tau \frac{1}{4}h(\tau)_{ij}\gamma_i\gamma_j}$ and we also define $B_{\mathcal{C}}(\tau_1,\tau_2) = \prod_{\tau=\tau_2}^{\tau_1} e^{-\Delta_\tau h(\tau)}$. For Majorana Green's functions $G_{ij} = \langle\gamma_i\gamma_j\rangle_{\mathcal{C}}$, the observable operator reads $O = \overline{\gamma}A^{(i,j)}\gamma$ with $A^{(i,j)}_{k,l} = \delta_{ki}\delta_{lj}$ and $0 \leqslant i,j < 2N$. We are going to prove that the finite-temperature Majorana Green's function is, however, free of the sign ambiguity that we stressed above, and the result is

$$G_{ij} = \langle\gamma_i\gamma_j\rangle_{\mathcal{C}} = 2\left[\mathbb{I} + B(\tau,0)B(\beta,\tau)\right]^{-1}_{ij}, \tag{11}$$

which shares a quite similar formula with the Green's function of complex fermions up to a prefactor of 2.

> **NOTE**: *For different conventions of Majorana transformation adopted, as in Eq.(2), the prefactor in Eq.(11) varies.*

To begin with, we need a few preparations on some basic facts about the skew-symmetric matrix and its Pfaffian.

> **PRELIMINARIES:**
>
> To obatin Eq.(11), we first introduce the following Pfaffian identity. For arbitrary $2n$-dimensional skew-symmetric matrix A, the pfaffian of A is given by
>
> $$\mathrm{Pf}(A) = i^{n^2} \exp\left(\frac{1}{2}\mathrm{Tr}\,\ln((\tau_y \otimes \mathbb{I}_{n\times n})^T \cdot A)\right), \tag{12}$$
>
> where τ_y denotes the 2×2 Pauli matrix. This equality is based on the trace identity $\mathrm{Pf}(A)\mathrm{Pf}(B) = \exp\left(\frac{1}{2}\mathrm{Tr}\,\ln(A^T B)\right)$ and on the observation that $\mathrm{Pf}(\tau_y \otimes \mathbb{I}_{n\times n}) = (-i)^{n^2}$.
>
> Another formula we need is that for invertible matrices A and B,
>
> $$\mathrm{Tr}\,\ln(AB) = \mathrm{Tr}\,\ln(A) + \mathrm{Tr}\,\ln(B) + i2\pi m, \quad m \in \mathbb{Z}. \tag{13}$$
>
> This can be argued from the BCH formula $e^X e^Y = e^{X+Y+C[X,Y]}$ by assuming $X = \ln A$ and $Y = \ln B$ and take logarithm and trace on the both sides. $C[X,Y]$ is a series of commutators of X and Y, and the trace of commutators definitely vanishes. The extra term of $i2\pi m$ is owing to the multivalued nature of matrix logarithm, e.g. $\ln e^A = A + i2\pi m\mathbb{I}$. Moreover, a stronger theorem exists which states that if A and B are both positive-definite matrices, then $m = 0$ and $\mathrm{Tr}\,\ln(AB) = \mathrm{Tr}\,\ln(A) + \mathrm{Tr}\,\ln(B)$.
>
> Below we show a python demo code for the numerical benchmarks of Eq.(13).
>
> ```python
> # test trace-logarithm of matrix products
> import numpy as np
> import scipy as sp
>
> shape = (10,10)
> A = np.random.rand(*shape) + 1.j * np.random.rand(*shape)
> B = np.random.rand(*shape) + 1.j * np.random.rand(*shape)
> trab = np.trace(sp.linalg.logm(A.dot(B)))
> tra = np.trace(sp.linalg.logm(A))
> trb = np.trace(sp.linalg.logm(B))
> print(trab, tra+trb, (tra+trb-trab)/(2*np.pi))
>
> Binv = np.linalg.inv(B)
> sqrtBinv = sp.linalg.sqrtm(Binv)
> trsqrtbinv = np.trace(sp.linalg.logm(sqrtBinv))
> print(trsqrtbinv, trb, 0.5*trb+trsqrtbinv)
> ```

We can then calculate the Majorana Green's function by resorting to the ancient trick of using generating functional,[3]

$$\langle O\rangle_{\mathcal{C}} = \frac{\partial}{\partial\eta}\ln\mathrm{Tr}\left[U_{\mathcal{C}}(\beta,\tau)e^{\eta O}U_{\mathcal{C}}(\tau,0)\right]\Big|_{\eta=0}. \tag{14}$$

To make use of our formula for the trace of exponential products, which is previously derived, the operator $O = \bar{\gamma} A^{(i,j)} \gamma$ shoulde be skew-symmetric, i.e. $A^{(i,j)} = -[A^{(i,j)}]^T$. Hence we turn to first consider $\langle O \rangle_{\mathcal{C}} = \langle [\gamma_i, \gamma_j] \rangle_{\mathcal{C}}$ with $A^{(i,j)}_{k,l} = \delta_{ki}\delta_{lj} - \delta_{kj}\delta_{li}$, and define $\tilde{B} = B_{\mathcal{C}}(\beta,\tau) e^{4\eta A} B_{\mathcal{C}}(\tau,0)$,

$$
\begin{aligned}
\langle [\gamma_i, \gamma_j] \rangle_{\mathcal{C}} &= \frac{\partial}{\partial \eta} \ln \mathrm{Tr}\left[U_{\mathcal{C}}(\beta,\tau) e^{\eta O} U_{\mathcal{C}}(\tau,0) \right]\Big|_{\eta=0} \\
&= \frac{\partial}{\partial \eta} \ln \left[\frac{\mathrm{Pf}(\tilde{B}^{-1} - \tilde{B})}{\mathrm{Pf}(\tilde{B}^{-1/2} - \tilde{B}^{1/2})} \right]\Bigg|_{\eta=0} \\
&= \frac{1}{2} \frac{\partial}{\partial \eta} \left[\mathrm{Tr}\ \ln(\tilde{B}^{-1} - \tilde{B}) - \mathrm{Tr}\ \ln(\tilde{B}^{-1/2} - \tilde{B}^{1/2}) + i2\pi m \right]\Big|_{\eta=0} \\
&= \frac{1}{2} \frac{\partial}{\partial \eta} \left[\mathrm{Tr}\ \ln(\tilde{B}^{-1/2} + \tilde{B}^{1/2}) + i2\pi m \right]\Big|_{\eta=0} \\
&= \frac{1}{2} \frac{\partial}{\partial \eta} \left[\mathrm{Tr}\ \ln(\mathbb{I} + \tilde{B}) - \frac{1}{2} \mathrm{Tr}\ \ln \tilde{B} + i2\pi m \right]\Big|_{\eta=0} \\
&= \frac{1}{2} \mathrm{Tr} \left[(\mathbb{I} - (\mathbb{I} + B_{\mathcal{C}}(\tau,0) B_{\mathcal{C}}(\beta,\tau))^{-1}) 4A^{(i,j)} - \frac{1}{2} 4A^{(i,j)} \right] \\
&= 2 \left\{ \left[\frac{1}{2}\mathbb{I} - (\mathbb{I} + B_{\mathcal{C}}(\tau,0) B_{\mathcal{C}}(\beta,\tau))^{-1}) \right]_{ji} - \left[\frac{1}{2}\mathbb{I} - (\mathbb{I} + B_{\mathcal{C}}(\tau,0) B_{\mathcal{C}}(\beta,\tau))^{-1}) \right]_{ij} \right\} \\
&= 2 \left([G^{cf}]_{ij} - [G^{cf}]_{ji} \right)
\end{aligned}
\tag{15}
$$

where we define $G^{cf} = (\mathbb{I} + B_{\mathcal{C}}(\tau,0) B_{\mathcal{C}}(\beta,\tau))^{-1}$ which is formally identical to the Green's function of complex fermions. In the 5th line, we assume that $m \in \mathbb{Z}$ does not depend on η, since the RHS should be an analytic function of η. (Otherwise the derivative over η blows up.) We can therefore ignore the $i2\pi m$ after taking the derivative of η as in the 6th line. We also note that using Eq.(9) to perform the trace over Hilbert space in step 2 leads to the identical result $\langle [\gamma_i, \gamma_j] \rangle_{\mathcal{C}} = 2\left([G^{cf}]_{ij} - [G^{cf}]_{ji} \right)$, which implies that there is no more sign ambiguity.

In order to obtain the Majorana Green's functions, we need a little more effort. First one can observe that $G^{cf} + [G^{cf}]^T = \mathbb{I}$.

PROVE: $G^{cf} + [G^{cf}]^T = \mathbb{I}$.

$$
\begin{aligned}
G^{cf} + [G^{cf}]^T &= (\mathbb{I} + B_{\mathcal{C}}(\tau,0) B_{\mathcal{C}}(\beta,\tau))^{-1} + [(\mathbb{I} + B_{\mathcal{C}}(\tau,0) B_{\mathcal{C}}(\beta,\tau))^{-1}]^T \\
&= (\mathbb{I} + B_{\mathcal{C}}(\tau,0) B_{\mathcal{C}}(\beta,\tau))^{-1} + \left(\mathbb{I} + B_{\mathcal{C}}(\beta,\tau)^T B_{\mathcal{C}}(\tau,0)^T\right)^{-1} \\
&= (\mathbb{I} + B_{\mathcal{C}}(\tau,0) B_{\mathcal{C}}(\beta,\tau))^{-1} + \left(\mathbb{I} + B_{\mathcal{C}}(\beta,\tau)^{-1} B_{\mathcal{C}}(\tau,0)^{-1}\right)^{-1} \\
&= (\mathbb{I} + B_{\mathcal{C}}(\tau,0) B_{\mathcal{C}}(\beta,\tau))^{-1} + (\mathbb{I} + B_{\mathcal{C}}(\tau,0) B_{\mathcal{C}}(\beta,\tau))^{-1} [B_{\mathcal{C}}(\beta,\tau)^{-1} B_{\mathcal{C}}(\tau,0)^{-1}]^{-1} \\
&= (\mathbb{I} + B_{\mathcal{C}}(\tau,0) B_{\mathcal{C}}(\beta,\tau))^{-1} (\mathbb{I} + B_{\mathcal{C}}(\tau,0) B_{\mathcal{C}}(\beta,\tau)) \\
&= \mathbb{I}.
\end{aligned}
\tag{16}
$$

> In the third step, we note that $B_{\mathcal{C}}(\tau_1,\tau_2)^T = [\prod_{\tau=\tau_1}^{\tau_2} e^{-\Delta_\tau h(\tau)}]^T = \prod_{\tau=\tau_2}^{\tau_1} e^{-\Delta_\tau h(\tau)^T} = \prod_{\tau=\tau_2}^{\tau_1} e^{\Delta_\tau h(\tau)} = B_{\mathcal{C}}(\tau_1,\tau_2)^{-1}$, where h is skew-symmetric, i.e. $h(\tau)^T = -h(\tau)$.

Hence the diagonal elements $[G^{cf}]_{ii} = \frac{1}{2}$ and furthermore the Green's function $\langle \gamma_i \gamma_i \rangle_{\mathcal{C}} = 1$ by definition. Combining these results, we can immediately write down

$$G_{ij} = \langle \gamma_i \gamma_j \rangle_{\mathcal{C}} = \begin{cases} \frac{1}{2}\langle [\gamma_i, \gamma_j] \rangle_{\mathcal{C}} = 2[G^{cf}]_{ij}, & i \neq j \\ 1, & i = j \end{cases} = 2\,[G^{cf}]_{ij} = 2\,[\mathbb{I} + B(\tau,0)B(\beta,\tau)]^{-1}. \tag{17}$$

Furthermore, it can be easily checked that using the definition of $G(\tau,\tau)$ above we can warp the equal-time Green's function at imaginary-time τ_1 to another time τ_2 in a quite straightfoward way, i.e.,

$$G(\tau_2,\tau_2) = B_{\mathcal{C}}(\tau_2,\tau_1) G(\tau_1,\tau_1) B_{\mathcal{C}}^{-1}(\tau_2,\tau_1). \tag{18}$$

This concludes our discussions on the equal-time Majorana Green's functions.

Time-displaced Majorana Green's functions

The time-displaced Majorana Green's function $\langle \gamma_i(\tau_1)\gamma_j(\tau_2)\rangle_{\mathcal{C}}$ is defined as $\langle \gamma_i(\tau_1)\gamma_j(\tau_2)\rangle_{\mathcal{C}}$. For $\tau_1 > \tau_2$, this reads

$$\begin{aligned} \langle \gamma_i(\tau_1)\gamma_j(\tau_2)\rangle_{\mathcal{C}} &= \frac{\mathrm{Tr}\left[U_{\mathcal{C}}(\beta,\tau_1)\gamma_i U_{\mathcal{C}}(\tau_1,\tau_2)\gamma_j U_{\mathcal{C}}(\tau_2,0)\right]}{\mathrm{Tr}\left[U_{\mathcal{C}}(\beta,0)\right]} \\ &= \frac{\mathrm{Tr}\left[U_{\mathcal{C}}(\beta,\tau_2) U_{\mathcal{C}}^{-1}(\tau_1,\tau_2)\gamma_i U_{\mathcal{C}}(\tau_1,\tau_2)\gamma_j U_{\mathcal{C}}(\tau_2,0)\right]}{\mathrm{Tr}\left[U_{\mathcal{C}}(\beta,0)\right]}, \end{aligned} \tag{19}$$

where we need to first evaluate $U_{\mathcal{C}}^{-1}(\tau_1,\tau_2)\gamma_i U_{\mathcal{C}}(\tau_1,\tau_2)$, which further boils down to computing $\gamma_i(\tau) = e^{\tau\frac{1}{4}\bar{\gamma}h\gamma}\gamma_i e^{-\tau\frac{1}{4}\bar{\gamma}h\gamma}$. Following the standard procedure here [3], we first differentiate $\gamma_i(\tau)$ with respect to τ, which is a artificial parameter but not the imaginary time, and this leads to

$$\begin{aligned} \frac{\partial \gamma_i(\tau)}{\partial \tau} &= e^{\tau\frac{1}{4}\bar{\gamma}h\gamma} \sum_{kl} \left[\frac{1}{4} h_{kl}\gamma_k\gamma_l, \gamma_i\right] e^{-\tau\frac{1}{4}\bar{\gamma}h\gamma} \\ &= e^{\tau\frac{1}{4}\bar{\gamma}h\gamma} \sum_{j} \left[\frac{1}{2}(h_{ji} - h_{ij})\gamma_j\right] e^{-\tau\frac{1}{4}\bar{\gamma}h\gamma} \\ &= -\sum_j h_{ij}\gamma_j(\tau) \quad \text{or} \quad \sum_j \gamma_j(\tau) h_{ji}, \end{aligned} \tag{20}$$

where in the last step we employ $h_{ij} = -h_{ji}$. Aftering integrating over τ and setting $\tau = 1$, without the loss of generality we can infer that

$$e^{\frac{1}{4}\bar{\gamma}h\gamma}\gamma_i e^{-\frac{1}{4}\bar{\gamma}h\gamma} = \sum_j [e^{-h}]_{ij}\gamma_j = \sum_j \gamma_j [e^{h}]_{ji}. \tag{21}$$

At the end of the day, by iteratively applying this formula, we obtain

$$
\begin{aligned}
U_{\mathcal{C}}^{-1}(\tau_1,\tau_2)\gamma_i U_{\mathcal{C}}(\tau_1,\tau_2) &= \prod_{\tau=\tau_1}^{\tau_2} e^{\Delta_\tau \frac{1}{4}\overline{\gamma}h(\tau)\gamma}\gamma_i \prod_{\tau=\tau_2}^{\tau_1} e^{-\Delta_\tau \frac{1}{4}\overline{\gamma}h(\tau)\gamma} \\
&= \sum_j \left[\prod_{\tau=\tau_2}^{\tau_1} e^{-\Delta_\tau h(\tau)}\right]_{ij}\gamma_j \quad \text{or} \quad \sum_j \gamma_j \left[\prod_{\tau=\tau_1}^{\tau_2} e^{\Delta_\tau h(\tau)}\right]_{ji} \\
&= \sum_j [B_{\mathcal{C}}(\tau_1,\tau_2)]_{ij}\gamma_j \quad \text{or} \quad \sum_j \gamma_j [B_{\mathcal{C}}^{-1}(\tau_1,\tau_2)]_{ji}.
\end{aligned} \tag{22}
$$

Inserting this into the definition of time-displaced Green's function, we get for $\tau_1 > \tau_2$

$$
\begin{aligned}
G(\tau_1,\tau_2)_{ij} &\equiv \langle \gamma_i(\tau_1)\gamma_j(\tau_2)\rangle_{\mathcal{C}} \\
&= \sum_k [B_{\mathcal{C}}(\tau_1,\tau_2)]_{ik}[G(\tau_2,\tau_2)]_{kj} \quad \text{or} \quad \sum_k [G(\tau_2,\tau_2)]_{kj}[B_{\mathcal{C}}^{-1}(\tau_1,\tau_2)]_{ki} \\
&= [B_{\mathcal{C}}(\tau_1,\tau_2)G(\tau_2,\tau_2)]_{ij} \quad \text{or} \quad [(2\mathbb{I}-G(\tau_2,\tau_2))B_{\mathcal{C}}^{-1}(\tau_1,\tau_2)]_{ji},
\end{aligned} \tag{23}
$$

where $G(\tau,\tau)$ denotes the equal-time Majorana Green's function at imaginary-time τ, which satisfies $G(\tau,\tau)+G^T(\tau,\tau)=2\mathbb{I}$. By noting that $B_{\mathcal{C}}^T = B_{\mathcal{C}}^{-1}$, one can find that the two expressions for $G(\tau_1,\tau_2)$ above are identical. Hence we can choose $G(\tau_1,\tau_2)=B_{\mathcal{C}}(\tau_1,\tau_2)G(\tau_2,\tau_2)$ in the case of $\tau_1>\tau_2$, which is exactly the form of time-displaced Green's function for complex fermions in AFQMC.

For $\tau_1<\tau_2$, $G(\tau_1,\tau_2)_{ij} = \langle\gamma_i(\tau_1)\gamma_j(\tau_2)\rangle_{\mathcal{C}} = -\langle\gamma_j(\tau_2)\gamma_i(\tau_1)\rangle_{\mathcal{C}} = -G(\tau_2,\tau_1)_{ji}$, which yields the following compact form of the Green's function

$$
G(\tau_1,\tau_2) = -G^T(\tau_2,\tau_1) = -[B_{\mathcal{C}}(\tau_2,\tau_1)G(\tau_1,\tau_1)]^T = -[2\mathbb{I}-G(\tau_1,\tau_1)]B_{\mathcal{C}}^{-1}(\tau_2,\tau_1). \tag{24}
$$

Rotating to the standard Hartree-Fock-Bogoliubov (HFB) basis

Although formulating the quantum Monte Carlo algorithm in the Majorana representation, in the end we need to evaluate the Green's functions of complex fermions, which turns out to be elementary physical degrees of freedom in most of quantum systems. Recall that the state vector in the HFB basis reads $c=(c_1,\ldots,c_N,c_1^\dagger,\ldots,c_N^\dagger)^T$, and in the Majorana basis $\gamma=(\gamma_1^1,\ldots,\gamma_N^1,\gamma_1^2,\ldots,\gamma_N^2)^T$. These two state vectors can be related to each other by

$$
c = U\gamma \equiv \frac{1}{2}\begin{pmatrix} \mathbb{I} & i\mathbb{I} \\ \mathbb{I} & -i\mathbb{I}\end{pmatrix}\gamma, \quad c^\dagger = \overline{\gamma}U^\dagger \equiv \frac{1}{2}\overline{\gamma}\begin{pmatrix} \mathbb{I} & \mathbb{I} \\ -i\mathbb{I} & i\mathbb{I}\end{pmatrix}. \tag{25}
$$

Denoting the Majorana Green's function as G^γ, then the Green's function G^c for complex fermions can be expressed as

$$
[G^c]_{ij} \equiv \langle c_i c_j^\dagger\rangle_{\mathcal{C}} = \langle [U\gamma]_i[\overline{\gamma}U^\dagger]_j\rangle_{\mathcal{C}} = \sum_{kl} U_{ik}\langle\gamma_k\gamma_l\rangle_{\mathcal{C}}U_{lj}^\dagger = [UG^\gamma U^\dagger]_{ij}, \tag{26}
$$

where the indices i,j run from 0 to $2N-1$.

For a generic BdG Hamiltonian in Eq.(1), we can formulate it in the HFB basis as

$$
H_{\text{BdG}} = \frac{1}{2}c^\dagger H c + \frac{1}{2}\text{Tr}[T]; \quad H = \begin{pmatrix} T & \Delta \\ -\Delta^* & -T^*\end{pmatrix}, \tag{27}
$$

and accordingly in the Majorana basis, (noting that $U^{-1}=2U^\dagger$)

$$H_{\mathrm{BdG}} = \frac{1}{2}\bar{\gamma}[U^\dagger H U]\gamma + \frac{1}{2}\mathrm{Tr}[T] = \frac{1}{4}\bar{\gamma}h\gamma + \frac{1}{2}\mathrm{Tr}[T]; \quad h \equiv U^{-1}HU. \tag{28}$$

Hence the Green's function G^c reads

$$G^c = \frac{1}{2}UG^\gamma U^{-1} = U\left[\mathbb{I} + \prod_n e^{-\Delta_\tau h(n)}\right]^{-1} U^{-1} = \left[\mathbb{I} + \prod_n e^{-\Delta_\tau U h(n) U^{-1}}\right]^{-1} = \left[\mathbb{I} + \prod_n e^{-\Delta_\tau H(n)}\right]^{-1}, \tag{29}$$

which is directly expressed in terms of Hamiltonian matrices in the HFB basis and shares the identical form as what we encounter in the standard AFQMC. For Hamiltonian with no pairing terms, e.g. $\Delta = 0$, we note that Eq.(29) degenerates to the Green's function in the usual Hartree-Fock basis.

It is also straightforward to check that in the HFB basis, the dynamic Green's functions $G^c(\tau, 0)$ / $G^c(0, \tau)$, together with their transformations when wrapped from τ_1 to τ_2, also have formally identical expressions as in AFQMC.

Below we provide a demo code to illustrate the transformation from G^γ to G^c, and check the canonical anti-commutation relations of complex fermions.

```
# transformation from Majorana Green's functions to the Green's functions of complex fermions
ns = 10
nt = 10
dt = 0.1
shape = (2*ns, 2*ns)
U = np.kron(np.array([[0.5, 0.5j],[0.5, -0.5j]]), np.identity(ns)) # transformation matrix
Bc = np.identity(shape[0])
Bm = np.identity(shape[0])
for t in range(nt):
    t = np.random.rand(ns,ns) + 1.j * np.random.rand(ns,ns)
    t = 0.5 * (t + np.conjugate(t).T)
    delta = np.random.rand(ns,ns) + 1.j * np.random.rand(ns,ns)
    delta = 0.5 * (delta - delta.T)
    H = np.zeros(shape, dtype=complex) # construct BdG hamiltonian
    H[:ns,:ns] = t
    H[ns:,ns:] = -np.conjugate(t)
    H[:ns,ns:] = delta
    H[ns:,:ns] = -np.conjugate(delta)
    h = 2*((np.conjugate(U).T).dot(H)).dot(U) # skew-symmetric and purely complex
    Bc = sp.linalg.expm(-dt*H).dot(Bc)
    Bm = sp.linalg.expm(-dt*h).dot(Bm)

# Majorana Green's function
gm = 2 * np.linalg.inv(np.identity(Bm.shape[0]) + Bm)
# Green's function of complex fermion <c c^+>
gc = np.linalg.inv(np.identity(Bc.shape[0]) + Bc)
print(abs(gc-(U.dot(gm)).dot(np.conjugate(U).T)).max())
print(np.trace(gc)/ns) # equal 1

gcc = np.zeros(shape, dtype=complex) # <c^+ c>
for i in range(2*ns):
    for j in range(2*ns):
        gcc[i][j] = gc[(i+ns)%(2*ns)][(j+ns)%(2*ns)]
```

```
# check the canonical anti-commutation relations of complex fermions
print(abs(gcc+gc.T-np.identity(gc.shape[0])).max()) # < c^+ c >_{ij} = [ 1 - <c c^+> ]_{ji}
```

References

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